

Comparative Analysis of Sweeping Frequency Jamming and Non-sweeping Frequency Jamming in Partial-band Jamming Based on Projectile-carried Communication Jamming

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Abstract—In this paper, based on the background of projectile-carried communication jamming, around the partial-band jamming mode, from the perspective of sweeping frequency jamming and non-sweeping frequency jamming, the two jamming modes were compared and analyzed by OPNET simulation software. The results showed that the bit error rate of sweeping frequency jamming and non-sweeping frequency jamming to enemy target was almost the same, and the corresponding jamming mode of projectile-carried communication should be selected according to the specific situation. The conclusion has a certain guiding significance for the design and development of projectile-carried communication jammer.

Keywords—projectile-carried communication jamming; partial-band jamming; sweeping frequency jamming; non-sweeping frequency jamming

I. INTRODUCTION

The difference between projectile-carried communication jamming and common communication jamming is that the jamming power required by projectile-carried environment is lower than that of common communication jamming, and the requirements for generation of jamming signal and reaction time are also relatively high. Therefore, in the projectile-carried communication jamming, we generally select partial-band jamming, which will save a part of the jamming power, so as to match the space on the projectile.

II. PARTIAL-BAND JAMMING

The projectile-carried communication jammer mainly jams the enemy frequency hopping radio [1][2][3]. There are three kinds of jamming methods in partial-band jamming for frequency hopping radio: the first is that the projectile-carried communication jammer uses the jamming signal with a certain proportion of bandwidth to jam the communication frequency band of frequency hopping radio, where a certain proportion refers to a part of the frequency hopping

communication band, but not all of it; the second is that the jamming signal frequency of the projectile-carried communication jammer scans the frequency in a certain way within a certain proportional of bandwidth, which is called sweeping frequency jamming; the third is multi-frequency continuous wave jamming [4].

III. COMMON POINTS AND DIFFERENCES BETWEEN SWEEPING FREQUENCY JAMMING AND NON-SWEEPING JAMMING IN PARTIAL-BAND JAMMING

A. Common Points

According to the analysis in the first section, the sweeping frequency jamming belongs to the partial-band jamming, and the non-sweeping frequency jamming (the first and the third kind) also belongs to the partial-band jamming.

B. Differences

The requirement of jamming power for sweeping frequency jamming mode is not very high. At the same time, different sweeping parameters can be set according to different targets to carry out targeted jamming [5]. What's more, the jamming signal frequency of sweeping frequency jamming is constantly changing, and the jumping law depends on the scanning mode of frequency.

IV. COMPARISON AND ANALYSIS OF JAMMING EFFECT OF SWEEPING FREQUENCY JAMMING AND NON-SWEEPING FREQUENCY JAMMING FOR FREQUENCY HOPPING RADIO IN PARTIAL-BAND JAMMING

It can be seen from the analysis in the first section that since sweeping frequency jamming and non-sweeping frequency jamming (the first and the third kind) belong to partial-band jamming, their jamming effects on FH radio will not be very different. We will discuss this problem from the perspective of simulation.

A. Selection of Simulation Tools

From the point of view of the constructability and operability of the simulation model, the editability and intuitiveness of the simulation results, we choose the network simulation software OPNET as the simulation tool [6][7].

B. Establishment of Projectile-carried Communication Jamming Model based on OPNET



Figure 1. Model of projectile-carried communication jamming system in OPNET environment

As shown in Fig. 1, the projectile-carried communication jammer is located in the center of the receiving end 3 of the shortwave frequency hopping radio and the receiving end 1 of the ultra-short wave frequency hopping radio.

1) Setting of simulation parameters

In order to achieve the purpose of comparing and analyzing the simulation results between sweeping frequency jamming and non-sweeping frequency jamming, this paper designed a projectile-carried communication jammer, which adopted two jamming modes, namely sweeping frequency jamming and non-sweeping frequency jamming, and the other parameters were consistent. The specific parameters were set as follows:

a) Short wave radio

- ① Frequency hopping band: 1.5MHz-30MHz;
- ② Frequency hopping bandwidth of short wave radio: 3.2KHz;
- ③ Information transmission rate of short wave radio: 2Mb/s [8];
- ④ Modulation mode of shortwave radio: 2FSK;
- ⑤ The transmitting power of short wave radio is 10W.

b) Ultra-short wave radio

- ① Frequency hopping band: 30MHz-300MHz;
- ② Frequency hopping bandwidth of ultra-short wave radio: 1600KHz;

- ③ Information transmission rate of ultra-short wave radio: 2Mb/s [8];

- ④ Modulation mode of ultra-short wave radio: 2FSK;

- ⑤ The transmitting power of ultra-short wave radio is 10W;

c) Projectile-carried communication jammer

- ① Information transmission rate: 2Mb/s;

- ② Transmission power: 20W;

- ③ Jamming mode: sweeping frequency jamming and pulse jamming [9];

2) Simulation results

a) Sweeping frequency jamming

- ① Throughput of whole network service

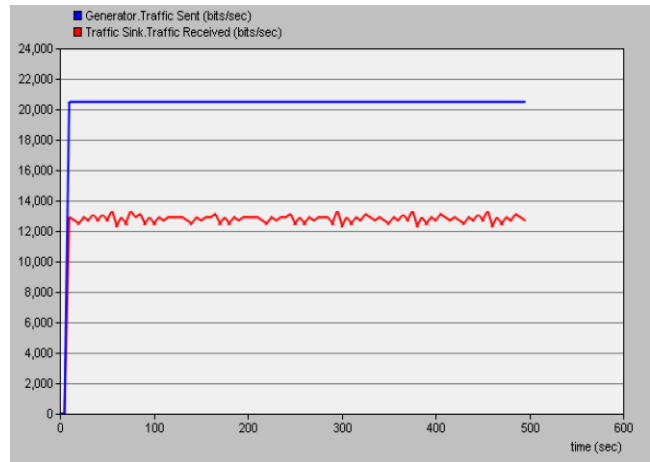


Figure 2. Throughput of whole network service

- ② End to end delay of whole network services

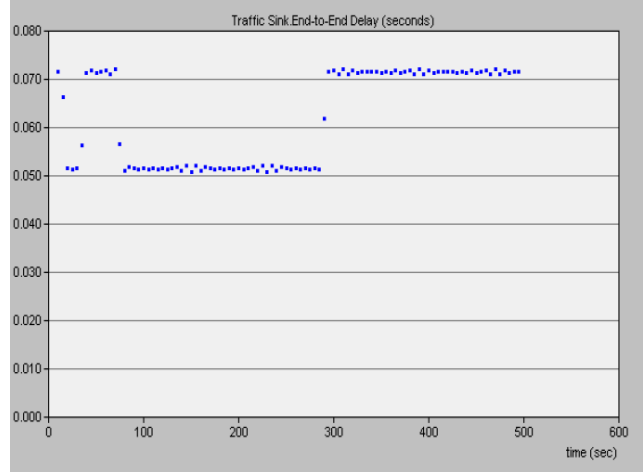


Figure 3. End to end delay of whole network services

- ③ Access delay of wireless channel in the whole network

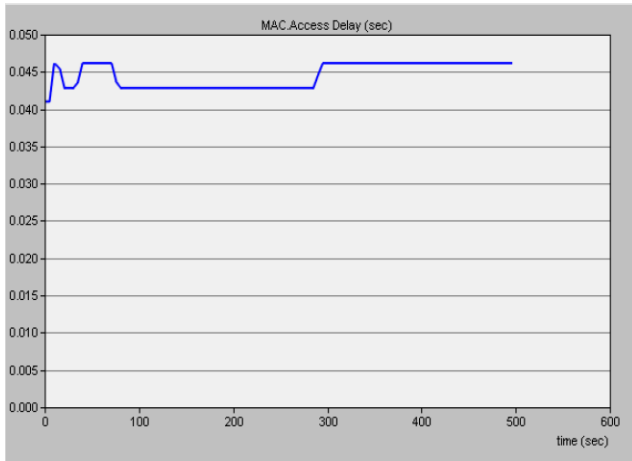


Figure 4. Access delay of wireless channel in the whole network

④ Bit error rate of radio model

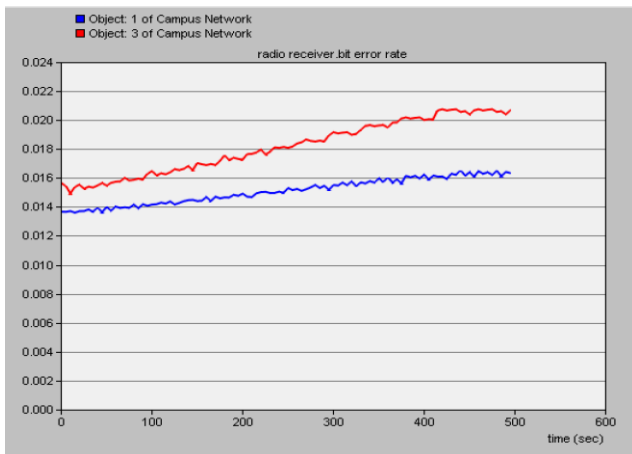


Figure 5. Bit error rate of radio model

⑤ Transmission throughput of jammer model

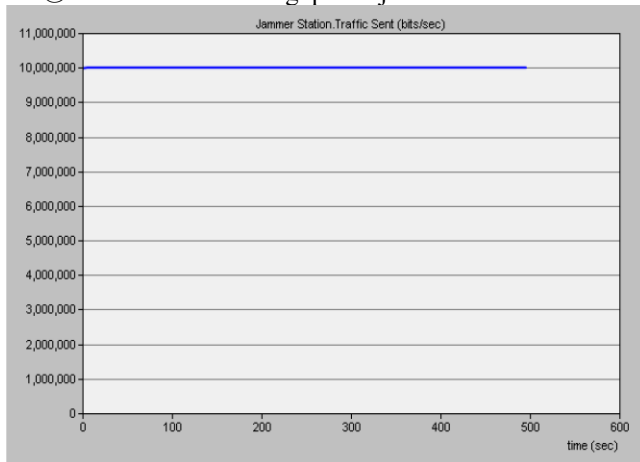


Figure 6. Transmission throughput of jammer model

b) Pulse jamming

① Throughput of whole network service

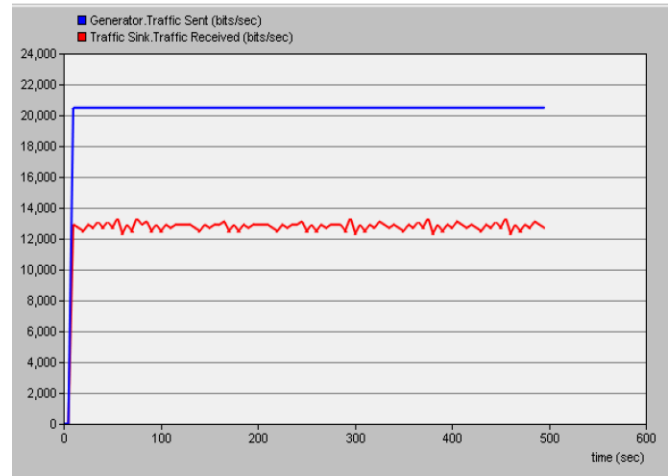


Figure 7. Throughput of whole network service

② End to end delay of whole network services

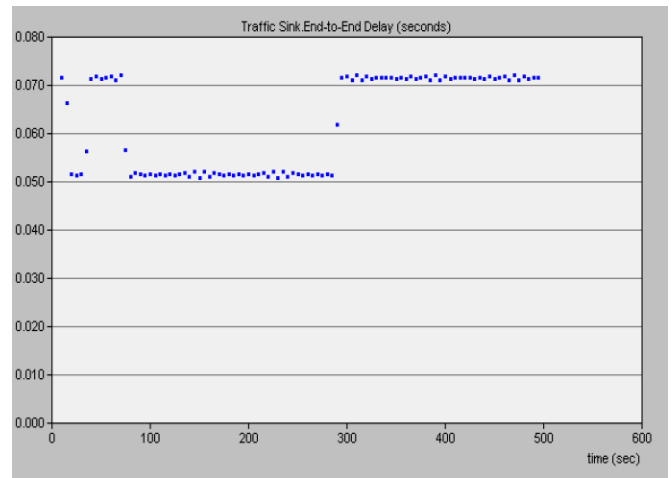


Figure 8. End to end delay of whole network services

③ Access delay of wireless channel in the whole network

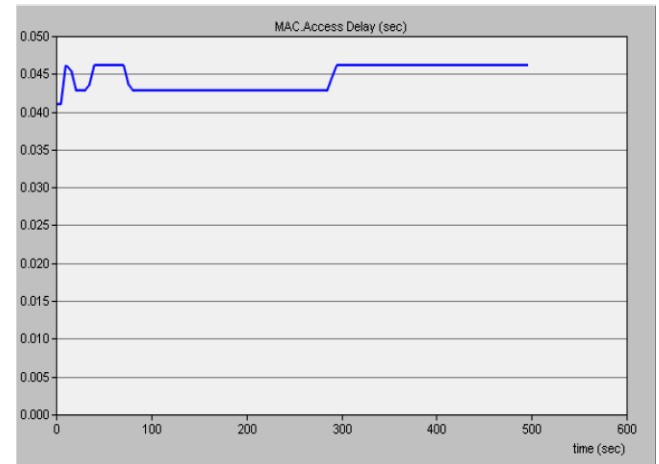


Figure 9. Access delay of wireless channel in the whole network

④ Bit error rate of radio model

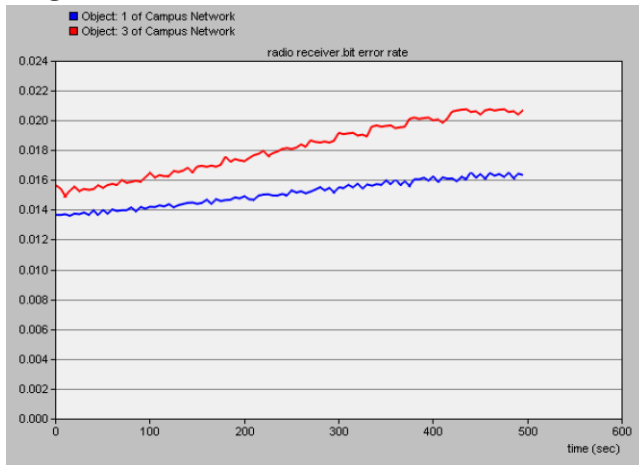


Figure 10. Bit error rate of radio model

⑤ Transmission throughput of jammer model

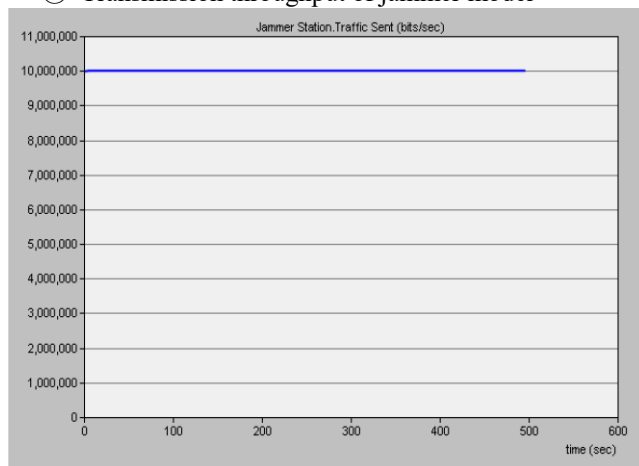


Figure 11. Transmission throughput of jammer model

3) Comparison of jamming effects of two jamming modes on frequency hopping radio

From the above simulation results, it could be seen that the bit error rate difference of the two jamming methods on FH radio was almost zero, whether it was sweep jamming or pulse jamming.

V. CONCLUSIONS

From the perspective of jamming effect, the sweeping frequency jamming and non-sweeping frequency jamming in partial-band jamming is almost the same. The difference is that sweeping frequency jamming is relatively more flexible, because it can change the relevant parameters and carry out more direct and effective jamming on the object to be jammed. Therefore, in practice, we should choose the corresponding partial-band jamming mode according to the specific situation. After all, the advantages and disadvantages of the jamming mode should be judged directly by the jamming effect.

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